

BASANTH NALLA

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Education

PDPM Indian Institute of Information Technology, Design and Manufacturing, Jabalpur Aug 2023 – Present

Bachelor of Technology (B.Tech), Computer Science and Engineering

Relevant Coursework: Data Structures & Algorithms, DBMS, Object-Oriented Programming, Computer Networks, Operating Systems, Computer Architecture, Operations Research, Data Science

Technical Skills

- **Programming Languages:** Python, JavaScript, C, C++, SQL
- **Backend Development:** Django REST Framework, Node.js, REST APIs, JWT Authentication, Role-Based Access Control, API Design
- **Frontend Development:** React.js, HTML5, CSS3
- **AI / LLM Technologies:** LangChain, RAG, Pinecone, Sentence Transformers, Prompt Engineering, Vector Databases, Large Language Models
- **Databases:** PostgreSQL, MySQL, SQLite, MongoDB, Query Optimization
- **Data Analysis:** Pandas, NumPy, Scikit-learn, Matplotlib, Seaborn, Excel, Jupyter Notebook, Statistical Analysis
- **Tools & Platforms:** Git, GitHub, GitHub Actions, AWS (EC2, S3 - Exposure)

Projects

Local-Konnect – Full-Stack Multi-Role Platform Jan 2025 – Jul 2025

React, Django REST Framework, PostgreSQL, JWT Authentication, REST APIs, Meta SAM

- Architected and created a full-stack platform serving customers, contractors, and supervisors with complete ownership of database schema, backend APIs, authentication, authorization, and frontend implementation.
- Built scalable Django REST APIs and PostgreSQL-backed workflows supporting tender creation, bidding, and contractor assignment processes.
- Integrated AI-powered interior visualization using Meta Segment Anything Model (SAM) for wall segmentation, color simulation, and texture recommendations.

AI-Powered Medical Chatbot – RAG + LLM System Oct 2025

Python, LangChain, Pinecone, Sentence Transformers, Groq LLaMA-3.3-70B

- Designed and implemented an end-to-end Retrieval-Augmented Generation pipeline including PDF ingestion, chunking, embedding generation, vector storage, semantic retrieval, and LLM response generation.
- Developed a domain-specific medical assistant capable of answering disease and symptom-related queries using curated medical knowledge sources.
- Built a multi-turn conversational interface that preserved context and improved user interaction quality.

Random Bakers – Full-Stack E-Commerce Web Application Oct 2024 – Dec 2024

React, Node.js, MySQL, REST APIs

- Made a complete e-commerce platform supporting authentication, product catalog browsing, cart management, and order processing.
- Used RESTful APIs enabling seamless communication between frontend and backend systems.
- Ensured secure session handling and transactional integrity across user workflows.

Retail Dataset EDA and Automation Scripting

Python, Pandas, NumPy, Matplotlib, Seaborn, Jupyter Notebook

- Automated cleaning and preprocessing of a retail dataset containing over 550,000 records.
- Performed exploratory data analysis to identify purchasing trends and key sales drivers.
- Developed reusable analytical workflows and visual reports for data-driven decision making.

Achievements

- Solved 150+ Data Structures & Algorithms problems on LeetCode and coding platforms.
- Maintained CGPA of 8.5/10 in B.Tech Computer Science curriculum.